

## Fourth moment of a characteristic polynomial in the style of Ingham

Ingham computed the fourth moment of  $\zeta$ :  $\frac{1}{T} \int_0^T |\zeta(1/2 + it)|^4 \sim \frac{1}{12} \zeta(2)^{-1} (\log T)^4$  as  $T \rightarrow \infty$ . The proof is too technical for the mini-course, and instead we explain what it looks like when applied to the unitary group  $U(N)$ . Let  $U_N$  be Haar-distributed on  $U(N)$ . We shall prove:

**Theorem 1.** *We have  $\mathbb{E}|\det(I - U_N)|^4 \sim \frac{N^4}{12}$  as  $N \rightarrow \infty$ .*

All the moments of  $\det(I - U_N)$  are known (see e.g. Keating–Snaith); the point of the argument below is to illustrate an argument originating in number theory.

*Proof.* The main idea is that the square  $\det(I - uU_N)^2$ , although it has degree  $2N$ , can be shortened to two pieces of length  $N$ , as we now explain. Recall we defined the secular coefficients of  $U_N$  via

$$\det(I - uU_N) = \sum_{n=0}^N (-1)^n \text{Sc}_n(U_N) u^n.$$

The functional equation for  $\det(I - uU_N)$  is

$$\det(I - uU_N) = (-u)^N \det U_N \overline{\det(I - \bar{u}^{-1}U_N)}. \quad (1)$$

We write

$$\det(I - uU_N)^2 = \sum_{q=0}^{2N} C_q(U_N) u^q$$

where

$$C_q(U) = (-1)^q \sum_{\substack{a+b=q \\ 0 \leq a, b \leq N}} \text{Sc}_a(U) \text{Sc}_b(U).$$

Squaring (1) gives

$$\det(I - uU_N)^2 = u^{2N} (\det U_N)^2 \overline{\det(I - \bar{u}^{-1}U_N)^2}.$$

Hence, by comparing coefficients,

$$C_{2N-q}(U_N) = (\det U_N)^2 \overline{C_q(U_N)}.$$

If we define the short polynomial

$$P_{U_N}(u) := \sum_{q=0}^{N-1} C_q(U_N) u^q,$$

then, for  $|u| = 1$ ,

$$\boxed{\det(I - uU_N)^2 = P_{U_N}(u) + u^{2N} (\det U_N)^2 \overline{P_{U_N}(\bar{u})} + C_N(U_N) u^N.} \quad (2)$$

Thus the degree- $2N$  polynomial  $\det(I - uU_N)^2$  is represented on the unit circle by two conjugate pieces of degree  $N - 1$ , plus the central coefficient  $C_N(U_N)$ . Taking  $u = 1$  in (2),

$$|\det(I - U_N)|^4 = \left| P_{U_N}(1) + (\det U_N)^2 \overline{P_{U_N}(1)} + C_N(U_N) \right|^2. \quad (3)$$

Expanding the square in (3), we have squared terms

$$|P_{U_N}(1)|^2, \quad |(\det U_N)^2 \overline{P_{U_N}(1)}|^2, \quad |C_N(U_N)|^2$$

and cross-terms

$$P_{U_N}(1) \overline{(\det U_N)^2 \overline{P_{U_N}(1)}}, \quad P_{U_N}(1) \overline{C_N(U_N)}, \quad (\det U_N)^2 \overline{P_{U_N}(1)} \overline{C_N(U_N)}$$

and their conjugates. After taking expectation, we claim that the cross-terms vanish. This is because each summand in the cross-terms is homogeneous of non-zero degree in the eigenvalues of  $U_N$ , which together with scalar invariance  $U_N \mapsto e^{i\phi} U_N$  of the Haar measure forces the expectation of the cross-terms to vanish. Consequently,

$$\mathbb{E}|\det(I - U_N)|^4 = 2\mathbb{E}|P_{U_N}(1)|^2 + \mathbb{E}|C_N(U_N)|^2.$$

Again by scalar invariance of the Haar measure,  $\mathbb{E}C_q(U_N) \overline{C_r(U_N)} = 0$  unless  $q = r$ . Hence

$$\mathbb{E}|P_{U_N}(1)|^2 = \mathbb{E} \left| \sum_{q=0}^{N-1} C_q(U_N) \right|^2 = \sum_{q=0}^{N-1} \mathbb{E}|C_q(U_N)|^2.$$

The following lemma is the remaining component.

**Lemma 1.** *For  $0 \leq q \leq N$ ,*

$$\mathbb{E}|C_q(U_N)|^2 = \binom{q+3}{3}.$$

Given the lemma,

$$\mathbb{E}|\det(I - U_N)|^4 = 2 \sum_{q=0}^{N-1} \binom{q+3}{3} + \binom{N+3}{3} = 2 \binom{N+3}{4} + \binom{N+3}{3} \sim \frac{N^4}{12}.$$

□

We do not include a proof of Lemma 1, but remark on why it is true: it is a general principle that if  $F(U_N)$  and  $G(U_N)$  are two homogeneous polynomials in the eigenvalues of  $U_N$ , whose degree is  $d \leq N$ , then  $\mathbb{E}F(U_N)\overline{G(U_N)}$  can be computed as follows: express  $F(U_N)$  and  $G(U_N)$  in terms of  $(\text{Tr}(U_N^i))_{i \leq d}$ , and then replace  $\text{Tr}(U_N^i)$  by  $N_i \sqrt{i}$  (where  $(N_i)_{i \leq d}$  are iid standard complex Gaussians). This is false if  $d = \deg F = \deg G$  exceeds  $N$ .

The simplest example is  $F(U_N) = G(U_N) = \text{Tr}(U_N^i)$  ( $i \leq N$ ) which was established by Diaconis and Shahshahani; in fact they consider products of such traces of degree at most  $N$ .

This principle is analogous to the fact that if  $M$  is a polynomial in  $\mathbb{F}_q[T]$ , then

$$\frac{1}{\phi(M)} \sum_{\chi \bmod M} \chi(f) \overline{\chi}(g)$$

is easy to compute if  $f, g$  are polynomials, coprime to  $M$ , of degree less than the degree of  $M$ . Indeed, in this range, the sum is 1 if  $f = g$  and 0 otherwise.